

Helix OTA — OpenDesign Token WCAG Contrast Audit

Revision: 1 **Last modified:** 2026-07-09T17:39:09Z **Scope:** design-systems/helix-ota/tokens.css (light :root + dark overrides) **Authority:** T1 review finding **M1** (dark --danger/--warn MAY fail WCAG) — verified here with MEASURED contrast ratios, not opinion. **Boundary (§11.4.6):** this is analysis only. No token was edited. The proposed fixes below are **PROPOSED (not applied)** — the tokens are vendored byte-identical into two frontends, so any change requires a coordinated re-vendor pass.

1. Method — the exact formula used

WCAG 2.1 relative luminance + contrast ratio (SC 1.4.3 / 1.4.11), computed by scratchpad/contrast.py (session scratchpad, NOT committed to the repo):

```
For each sRGB 8-bit channel value c (0..255):
    c' = c / 255
    linear(c) = c' / 12.92                if c' <= 0.03928
                ((c' + 0.055) / 1.055) ^ 2.4 otherwise
Relative luminance:
    L = 0.2126 * R_lin + 0.7152 * G_lin + 0.0722 * B_lin
Contrast ratio (L1 = lighter, L2 = darker of the two):
    ratio = (L1 + 0.05) / (L2 + 0.05)
```

Thresholds applied. AA normal text = **4.5:1** (SC 1.4.3). AA large text (≥18pt / ≥14pt-bold) and non-text UI components / graphical objects = **3:1** (SC 1.4.3 large + SC 1.4.11 non-text). Each pair states which threshold applies and why.

Every ratio below is a real computed number — the literal script output is pasted in §2 and §5.

2. Literal script output (measured)

PAIR	THRESH	LIGHT	DARK
note			

--fg on --bg	4.5	20.01 PASS	19.12 PASS
L:#020817on#ffffff D:#f8fafcon#020817			
--fg on --surface	4.5	20.01 PASS	19.12 PASS
L:#020817on#ffffff D:#f8fafcon#020817			
--fg on --surface-warm	4.5	18.26 PASS	13.98 PASS
L:#020817on#f1f5f9 D:#f8fafcon#1e293b			
--muted on --bg	4.5	4.76 PASS	7.80 PASS
L:#64748bon#ffffff D:#94a3b8on#020817			
--muted on --surface	4.5	4.76 PASS	7.80 PASS
L:#64748bon#ffffff D:#94a3b8on#020817			
--muted on --surface-warm	4.5	4.34 FAIL	5.71 PASS
L:#64748bon#f1f5f9 D:#94a3b8on#1e293b			
--accent on --bg	3.0	5.17 PASS	5.44 PASS
L:#2563ebon#ffffff D:#3b82f6on#020817			
--accent-on on --accent	4.5	4.94 PASS	4.85 PASS
L:#f8fafcon#2563eb D:#0f172aon#3b82f6			
--danger on --bg	3.0	4.83 PASS	2.00 FAIL
L:#dc2626on#ffffff D:#7f1d1don#020817			
--danger on --surface	3.0	4.83 PASS	2.00 FAIL
L:#dc2626on#ffffff D:#7f1d1don#020817			
--danger on --surface-warm	3.0	4.41 PASS	1.46 FAIL
L:#dc2626on#f1f5f9 D:#7f1d1don#1e293b			
--accent-on on --danger	4.5	4.62 PASS	1.78 FAIL
L:#f8fafcon#dc2626 D:#0f172aon#7f1d1d			
--success on --bg	3.0	3.30 PASS	6.07 PASS
L:#16a34aon#ffffff D:#16a34aon#020817			
--success on --surface-warm	3.0	3.01 PASS	4.44 PASS
L:#16a34aon#f1f5f9 D:#16a34aon#1e293b			
--warn on --bg	3.0	1.92 FAIL	10.43 PASS
L:#eab308on#ffffff D:#eab308on#020817			
--warn on --surface-warm	3.0	1.75 FAIL	7.63 PASS
L:#eab308on#f1f5f9 D:#eab308on#1e293b			
--border on --surface	3.0	1.23 FAIL	1.37 FAIL
L:#e2e8f0on#ffffff D:#1e293bon#020817			
--border on --bg	3.0	1.23 FAIL	1.37 FAIL
L:#e2e8f0on#ffffff D:#1e293bon#020817			
--border on --surface-warm	3.0	1.13 FAIL	1.00 FAIL
L:#e2e8f0on#f1f5f9 D:#1e293bon#1e293b			

```

=== FLAGGED DARK SEMANTICS (M1) detail ===
--danger(dark #7f1d1d) on --bg(#020817): 1.997:1
--danger(dark #7f1d1d) on --surface(#020817): 1.997:1
--warn(inherited #eab308) on --bg dark(#020817): 10.432:1
--accent-on(dark #0f172a) on --danger(dark #7f1d1d): 1.782:1
(dark on dark)
--fg(dark #f8fafc) on --danger(dark #7f1d1d): 9.576:1 (light
label on danger fill)

```

3. Per-pair measured results

Threshold column: **4.5** = normal text (SC 1.4.3); **3.0** = UI component / large text (SC 1.4.11 / 1.4.3-large).

Pair (foreground on background)	Light hex	Light ratio	Light verdict	Dark hex	Dark ratio	Dark verdict	Threshold
--fg on --bg	#020817/ #ffffff → #f8fafc/ #020817	20.01	PASS	—	19.12	PASS	4.5 (body text)
--fg on -- surface	same as bg	20.01	PASS	—	19.12	PASS	4.5 (body text)
--fg on -- surface-warm	#020817/ #f1f5f9 → #f8fafc/ #1e293b	18.26	PASS	—	13.98	PASS	4.5 (body text on panel)
--muted on -- bg	#64748b/ #ffffff → #94a3b8/ #020817	4.76	PASS	—	7.80	PASS	4.5 (secondary text)
--muted on -- surface	same as bg	4.76	PASS	—	7.80	PASS	4.5 (secondary text)
--muted on -- surface-warm	#64748b/ #f1f5f9 →	4.34	FAIL	—	5.71	PASS	4.5 (secondary

Pair (foreground on background)	Light hex	Light ratio	Light verdict	Dark hex	Dark ratio	Dark verdict	Threshold
	#94a3b8/ #1e293b						text on panel)
--accent on --bg	#2563eb/ #ffffff → #3b82f6/ #020817	5.17	PASS	—	5.44	PASS	3.0 (link/UI accent) ¹
--accent-on on --accent	#f8fafc/ #2563eb → #0f172a/ #3b82f6	4.94	PASS	—	4.85	PASS	4.5 (button label on fill)
--danger on --bg	#dc2626/ #ffffff → #7f1d1d/ #020817	4.83	PASS	—	2.00	FAIL	3.0 (status/ icon) ²
--danger on --surface	same as bg	4.83	PASS	—	2.00	FAIL	3.0 (status/ icon) ²
--danger on --surface- warm	#dc2626/ #f1f5f9 → #7f1d1d/ #1e293b	4.41	PASS	—	1.46	FAIL	3.0 (status/ icon) ²
--accent-on on --danger (danger-as- fill label)	#f8fafc/ #dc2626 → #0f172a/ #7f1d1d	4.62	PASS	—	1.78	FAIL	4.5 (label on fill) ³
--success on --bg	#16a34a/ #ffffff → #16a34a/ #020817	3.30	PASS	—	6.07	PASS	3.0 (status/ icon) ⁴
--success on --surface- warm	#16a34a/ #f1f5f9 →	3.01	PASS	—	4.44	PASS	3.0 (status/ icon)

Pair (foreground on background)	Light hex	Light ratio	Light verdict	Dark hex	Dark ratio	Dark verdict	Threshold
--warn on -- bg	#16a34a/ #1e293b #eab308/ #ffffff → #eab308/ #020817	1.92	FAIL	—	10.43	PASS	3.0 (status/ icon) ⁵
--warn on -- surface-warm	#eab308/ #f1f5f9 → #eab308/ #1e293b	1.75	FAIL	—	7.63	PASS	3.0 (status/ icon) ⁵
--border on --surface	#e2e8f0/ #ffffff → #1e293b/ #020817	1.23	FAIL	—	1.37	FAIL	3.0 (non- text boundary) ⁶
--border on --bg	same as surface	1.23	FAIL	—	1.37	FAIL	3.0 (non- text boundary) ⁶
--border on --surface- warm	#e2e8f0/ #f1f5f9 → #1e293b/ #1e293b	1.13	FAIL	—	1.00	FAIL	3.0 (non- text boundary) ⁶

Footnotes: 1. --accent is measured at 3:1 (its role is links / UI accent / focus). If accent is ever used as **normal-size accent-colored body text**, the 4.5 bar applies — light 5.17 and dark 5.44 clear that too, so accent passes either way. 2. Dark --danger #7f1d1d is a dark maroon. As a **foreground tone** (danger text, icon, badge outline, chart series) on the dark background it measures **2.00:1** (bg/surface) and **1.46:1** (warm panel) — below the 3:1 UI floor and far below 4.5 for text. **This is finding M1, CONFIRMED for --danger (dark)**. 3. If dark --danger is instead used as a **filled** surface, the on-color pairing in the tokens (--accent-on = dark #0f172a) gives **1.78:1** — dark-on-dark, also failing. A light label (--fg #f8fafc) on #7f1d1d measures 9.58:1 (passes), so today's dark danger fill is only legible with a LIGHT label, which the token

pairing does not provide. Either way the current dark danger token is broken for at least one of its two roles. 4. Light --success clears 3:1 but is tight on the warm panel (3.01) — acceptable for UI/icon use, would FAIL 4.5 if used as success **body text** (compute: $3.30 < 4.5$). Flag for text use, not a hard fail at the UI threshold. 5. **M1 for --warn is NOT confirmed in the direction flagged.** Dark --warn inherits the light value #eab308 (the dark block does not override it) and measures **10.43 / 7.63** on dark surfaces — comfortably PASS. It is the **LIGHT** theme where --warn #eab308 (bright yellow) FAILS on white: **1.92 / 1.75**. So warn has a real contrast defect, but in light mode, not dark. 6. --border fails 3:1 on every surface in both themes. **Nuance (SC 1.4.11):** the 3:1 non-text requirement applies only to borders that are the **sole** visual means of identifying a control (e.g. an un-filled input's outline). Purely **decorative** dividers / card outlines are exempt and are NOT a violation. So these rows are “FAIL *if* the border is load-bearing for a control boundary” — see §4 recommendation 4 for the surgical fix (a distinct strong-border token) rather than darkening every divider.

Tokens not cleanly computable (handled honestly, §11.4.6)

- --accent-hover = color-mix(in oklab, var(--accent), black 8%), --accent-active = ... black 14% — resolve to a **darker** accent. On a light background darker = higher contrast, so if --accent passes (it does), these pass by construction; on a dark background they reduce contrast slightly but start from 5.44 with an 8–14% black mix. Exact hex depends on the browser's oklab mix; not a single fixed sRGB value. Marked **UNKNOWN (bounded — not independently failing)**.
- --focus-ring = color-mix(..., var(--accent), transparent 70%), --elev-raised / --elev-ring use transparent/alpha — these composite over whatever is behind them, so their effective contrast depends on the backdrop pixel and cannot be reduced to one ratio. Marked **UNKNOWN: alpha-composited, needs per-context render sampling**.

4. Findings summary

#	Token (theme)	Measured	Threshold	Verdict	Note
M1a		2.00 / 1.46	3.0	FAIL	

#	Token (theme)	Measured	Threshold	Verdict	Note
	--danger (DARK) as foreground				Confirmed T1 M1. Dark maroon unreadable on dark surfaces.
M1b	--danger (DARK) as fill + token on-color	1.78	4.5	FAIL	Paired on- color is dark-on- dark.
M1c	--warn — real defect is LIGHT not dark	L 1.92/1.75 · D 10.43/7.63	3.0	FAIL (light only)	Flag direction corrected: dark warn PASSES.
F4	--muted on --surface- warm (LIGHT)	4.34	4.5	FAIL	0.16 short; only on the warm panel.
F5	--border all surfaces (both themes)	1.00–1.37	3.0	FAIL (if load- bearing)	Exempt when decorative; fix control boundaries only.
—	--success (LIGHT) as body text	3.30	4.5	note	Passes 3:1 UI; fails 4.5 text. Restrict to UI/icon use.
—	all --fg, -- accent, -- accent-on, --muted(bg/ surface)	≥4.76	—	PASS	Core text + accent are solid in both themes.

5. PROPOSED fixes — NOT APPLIED

PROPOSED (not applied — coordinated re-vendor required since two frontends vendor tokens.css byte-identical). Do NOT edit tokens.css from this audit. Every candidate ratio below is measured (scratchpad/propose.py output pasted after the table). Candidates are chosen from the same Tailwind palette family the tokens are already sourced from, to stay on-brand.

Finding	Token / theme	Current	Current ratio	PROPOSED	New ratio (measured)	Now meets
M1a/ M1b	--danger DARK	#7f1d1d	2.00 bg · 1.46 warm	#dc2626 (Tailwind red-600 — unifies with light)	4.14 bg · 3.03 warm; light-label #f8fafc fill = 4.62	3:1 foreground both surfaces and 4.5 fill- label
M1a (higher margin alt)	--danger DARK	#7f1d1d	—	#ef4444 (red-500)	5.32 bg · 3.89 warm	stronger 3:1; but fill- label #f8fafc = 3.60 (<4.5) so fill needs dark label
M1c	--warn LIGHT	#eab308	1.92 / 1.75	#a16207 (amber-700)	4.92 white · 4.49 warm	4.5 text on white (warm 4.49 ≈ AA, treat ≥3 UI as safe)
F4	--muted LIGHT	#64748b	4.34 warm	#475569 (slate-600)	6.92 warm · 7.58 bg	4.5 on all light surfaces w/ margin
F4 (minimal alt)	--muted LIGHT	#64748b	4.34 warm	#556070 (custom)	5.82 warm	minimal nudge, off- Tailwind
F5	new -- border- strong (both themes, for	n/a	—	light #64748b, dark #64748b	L 4.76 bg · 4.34 warm; D 4.20 bg · 3.07 warm	3:1 SC 1.4.11 for load- bearing boundaries

Finding	Token / theme	Current	Current ratio	PROPOSED	New ratio (measured)	Now meets
	control boundaries)					keep base --border for decorative dividers

Recommended primary set: dark --danger → **#dc2626** (single value satisfies both foreground-3:1 and fill-label-4.5, and unifies danger across themes); light --warn → **#a16207**; light --muted → **#475569**; add --**border-strong #64748b** rather than darkening the base border. Warn/danger fills that carry a label must pair the correct on-color — verify the chosen label in the re-vendor pass (not re-checked here beyond the values shown).

LIGHT --muted candidates (target >=4.5 on surface-warm #f1f5f9)

#64748b	-> bg#fff=4.76	warm#f1f5f9=4.34	(current)
#556070	-> bg#fff=6.38	warm#f1f5f9=5.82	
#475569	-> bg#fff=7.58	warm#f1f5f9=6.92	(slate-600)

LIGHT --warn candidates (target >=4.5 text on white / >=3 UI)

#eab308	-> bg#fff=1.92	warm#f1f5f9=1.75	(current, amber-500)
#ca8a04	-> bg#fff=2.94	warm#f1f5f9=2.68	(amber-600 – still <3)
#a16207	-> bg#fff=4.92	warm#f1f5f9=4.49	(amber-700)
#854d0e	-> bg#fff=6.85	warm#f1f5f9=6.25	(amber-800)

DARK --danger candidates (target >=3 on bg #020817; >=4.5 if body text)

#7f1d1d	-> bg#020817=2.00	warm#1e293b=1.46	(current, red-900)
#dc2626	-> bg#020817=4.14	warm#1e293b=3.03	(red-600)
#ef4444	-> bg#020817=5.32	warm#1e293b=3.89	(red-500)
#f87171	-> bg#020817=7.23	warm#1e293b=5.29	(red-400)

DARK danger-as-fill: light label #f8fafc on lightened danger
on #dc2626: 4.62 on #ef4444: 3.60 on #b91c1c: 6.18

BORDER --border-strong candidates (target >=3 vs surface)

LIGHT	on #ffffff: #94a3b8=2.56	#64748b=4.76
DARK	on #020817: #475569=2.64	#64748b=4.20 (warm #1e293b: 3.07)

6. Honest boundary

This audit proves per-pair **contrast** against WCAG AA — it does not assert overall WCAG conformance (which also depends on real component composition, text sizes at render time, disabled-state and placeholder rules, and the alpha-composited tokens marked UNKNOWN above). The proposed hex values are measured but **not applied**; applying them requires the coordinated re-vendor of `tokens.css` into both frontends so the two copies stay byte-identical.

Sources verified

Sources verified 2026-07-09: - WCAG 2.1 SC 1.4.3 Contrast (Minimum) — <https://www.w3.org/WAI/WCAG21/Understanding/contrast-minimum.html> - WCAG 2.1 SC 1.4.11 Non-text Contrast — <https://www.w3.org/WAI/WCAG21/Understanding/non-text-contrast.html> - WCAG relative luminance + contrast-ratio definitions — <https://www.w3.org/TR/WCAG21/#dfn-relative-luminance> and <https://www.w3.org/TR/WCAG21/#dfn-contrast-ratio> - Audited token file: `design-systems/helix-ota/tokens.css` (`light :root + @media (prefers-color-scheme: dark) / :root[data-theme="dark"] / .dark`) - Computation scripts (session scratchpad, NOT committed): `contrast.py`, `propose.py`